Determination of Seidel Aberration Coefficients in Scanning-Electron Microscopy Using Through-Focus Probe Shapes

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(Dated: January 12, 2026)

Scanning electron microscopes operate with electrons accelerated to high speeds, corresponding to low wavelengths. At such low wavelengths, the resolution is limited by the size of the electron probe rather than diffraction. The primary cause for a larger probe are aberrations produced by the electromagnetic lenses, in particular spherical aberrations. This paper attempts to quantify Seidel aberrations in scanning electron microscopes by generating point spread functions that fit extracted probe shapes. The probe shapes are extracted from images of tin balls on carbon with controlled defocus values. Underfocused images tend to be better fit than overfocused images of the same aberrations. This is potentially due to artefacts on the edge of the generated point spread function that are not present on the extracted point spread function. With a large database of aberrations and respective point spread functions, it could be possible to use machine learning to develop faster and more precise fitting algorithms.

I. INTRODUCTION

Traditional optical microscopes, while highly versatile, are fundamentally limited in resolution by diffraction [1], which is determined by the wavelength of visible light they use. In contrast, scanning electron microscopes (SEMs) use electrons instead of photons. The wavelength of these electrons depends on their velocity, which is controlled by accelerating them through large electric potential differences [2]. Because the resulting electron wavelengths are much shorter than those of visible light, diffraction ceases to be the primary factor limiting SEM resolution.

The resolution of SEMs is constrained by the size of the electron probe [3]. It is primarily affected by aberrations introduced by the electron lenses that focus the beam [4]. Among these, spherical aberrations are unavoidable in circular lenses, limiting achievable resolution to values up to 100 times the electrons' wavelength [3–5]. Over the past few years, significant research has focused on methods to automatically correct these aberrations [6, 7]. However, there is little research on quantifying the changes in aberration intensity following the use of aberration correctors in the microscope.

This paper investigates the feasibility and accuracy of quantifying Seidel aberrations by fitting generated point spread functions (PSFs) to probe shapes extracted from SEM images. The probe shapes are obtained from images of tin balls captured at a series of controlled defocus values, including both under- and over-focus. While the program fits several Seidel aberrations, particular emphasis is placed on spherical aberrations, as they are prominent in SEM imaging [3]. In section II, wavefront optics are outlined, including a summary of the aberrations considered in this study. section III details the process of probe shape extraction, PSF generation, and the subsequent fitting of aberration coefficients. The results, including verification and evaluation of the aberration extraction program, are presented in section IV and discussed in section V.

II. THEORY

A. Scanning Electron Microscope

In scanning electron microscopy, an image is formed by raster-scanning a focused electron probe across the specimen surface while recording the emitted signal intensity [8]. The probe originates from an electron source and is subsequently shaped and controlled as it travels down the electron column. After extraction, the electrons pass through a series of electromagnetic condenser lenses, as shown in Figure 1, which generate magnetic fields that exert a Lorentz force on the electrons and steer their trajectories toward the optical axis. By adjusting the lens coil currents or modifying the pole pieces, the magnetic field strength can be varied, allowing control over the beam diameter, convergence, and focal properties [2, 8].

To further refine the beam, one or more apertures are placed along the column to limit off-axis electrons and regulate the probe current, thereby improving beam coherence and resolution. The conditioned electron beam is then focused by the objective lens into a small, well-defined probe on the specimen surface. Scanning coils positioned above the objective lens deflect the probe in a controlled raster pattern, systematically sweeping it across the area of interest and enabling the point-by-point acquisition of image information [8].

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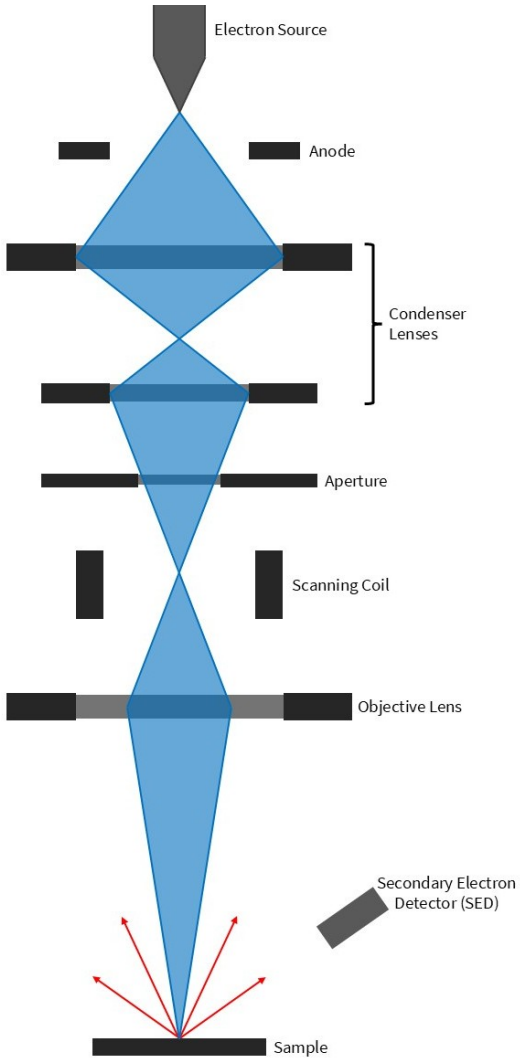


FIG. 1: General schematic of a SEM showing the electron source, condenser lenses, apertures, scanning coils, and objective lens. The structure and positioning of elements may vary between instruments.

The main difficulty in forming an ideal electron probe is that electromagnetic lenses do not offer the same design freedom as conventional glass lenses. In light optics, aberrations can be strongly reduced by shaping lens surfaces and by combining multiple converging and diverging elements with different refractive properties [9]. In contrast, the shapes of magnetic and electrostatic electron lenses cannot be freely designed. Moreover, they are inherently positive-power (converging) elements, preventing the use of compensating defocusing lenses commonly employed in light optics. This restriction arises because the electromagnetic potentials in the lens must satisfy Laplace's equation in regions free of charge, which fundamentally limits the possible field distributions [10].

Even for perfectly aligned, space-charge-free lenses, intrinsic aberrations such as spherical and chromatic aberration are unavoidable, as formalized by Scherzer's theorem [5]. These aberrations primarily result from the rotational symmetry of typical electron lenses: the outer zones of the lens focus electrons more strongly than the inner zones, producing spherical aberration, while variations in electron velocity cause different focal strengths, leading to chromatic aberration. Such fundamental limitations necessitate careful characterization and modeling of lens aberrations in electron microscopy.

B. Wavefront Optics

1. *Electrons as Waves*

The De Broglie wavelength of an electron is directly related to its momentum. An electron in an SEM microscope is accelerated to a high (relativistic) velocity through a certain potential U. Taking into account relativistic effects, the De Broglie wavelength λ of an electron accelerated by a potential ΔU in a vacuum is given by

$$\lambda = \frac{h}{\sqrt{2m_0 q\Delta U + \frac{1}{c^2}(q\Delta U)^2}}, \tag{1}$$

where h is Planck's constant, m_0 is the stationary mass of an electron, c is the speed of light and q is the charge of an electron.

2. *Point Spread Function*

The way an image is produced by an optical system can be defined using a point spread function (PSF). The intensity point spread function $\Psi(x, y)$, when convolved with the intrinsic signal $s(x, y)$, results in the image $i(x, y)$ produced by the optical system. This is shown in Equation 2 [11].

$$i(x, y) = \Psi(x, y) * s(x, y) \tag{2}$$

For an ideal system without any aberrations, the point spread function consists of a centered Dirac delta function, with each point in the produced image corresponding exactly to the sample [11]. However, the magnetic lenses in scanning electron microscopes are unable to focus the electron beam into a perfect singular beam, resulting in a aberrated point spread functions which produce images containing various optical artefacts such as defocus, astigmatism, coma, and spherical aberration. This is expanded upon in section II B 4.

3. *Fourier Optics*

It is possible to describe an optical system (and hence the point spread function) using Fourier optics [11]. In

Fourier optics, the pupil function $P(x', y')$ defines how the wavefront changes after passing through an aperture and a lens, and is defined as

$$P(x', y') = A(x', y')e^{j\frac{2\pi}{\lambda}\phi(x', y')} \tag{3}$$

where

$$A(x', y') = \begin{cases} 1, & \text{if } \sqrt{x'^2 + y'^2} \leq \theta \\ 0, & \text{if } \sqrt{x'^2 + y'^2} > \theta \end{cases} \tag{4}$$

for a circular aperture with maximum ray angle θ. The variables x' and y' refer to the angle of rays passing through the aperture, instead of the position. The wavelength λ and $\phi(x', y')$ defines the phase change that waves experience after passing through the lens [11].

Assuming a paraxial wavefront, the normalised complex point spread function $U(x, y)$ can be obtained through the Fourier transform of the pupil function given by Equation 5 [7, 11].

$$U(x, y) = \frac{1}{\pi\theta^2} \iint_{-\infty}^{\infty} P(x', y')e^{-j\frac{2\pi}{\lambda}(xx'+yy')}dx'dy'. \tag{5}$$

The variables x and y are position coordinates on the image plane. The intensity point spread function $\Psi(x, y)$ is obtained by squaring the magnitude of $U(x, y)$.

4. Wavefront Aberrations

Aberrations can be quantified using the aberrated phase function $\phi(x', y')$, which forms part of the overall pupil equation $P(x', y')$. It is made up of individual phase changes, which are contributed by different aberration modes. The aberration modes and respective phase contributions considered in this paper are summarised in Table I.

TABLE I: Seidel aberration modes and their contributions to the aberrated phase function $\phi(x', y')$ [7, 12], where x' and y' denote pupil angular coordinates, and θ is the maximum beam angle transmitted through the aperture.

Aberration	**Coefficient**	$\boldsymbol{\Delta\phi(x', y')}$
Defocus	C_{df}	$\frac{1}{2}C_{df}(x'^2 + y'^2)$
Two-fold Astigmatism	A_1	$Re\left[\frac{1}{2}A_1(x' - iy')^2\right]$
Three-fold Astigmatism	A_2	$Re\left[\frac{1}{3}A_2(x' - iy')^3\right]$
Coma	B_2	$Re\left[\bar{B}_2(x' - iy')\left(x'^2 + y'^2 - \frac{2}{3}\theta^2\right)\right]$
Spherical	C_s	$Re\left[\frac{1}{4}C_s(x'^2 + y'^2)(x'^2 + y'^2 - \theta^2)\right]$

Defocus is quantified by the real coefficient C_{df}, which characterizes the magnitude of the quadratic phase aberration. Its sign determines the direction of defocus, corresponding to a focal shift either in front of or behind the image plane.

Astigmatism is an aberration in which optical rays along different axes focus at different distances [11]. As a result, the point spread function becomes elongated along specific axes, depending on the type of astigmatism and the focus.

Two-fold astigmatism is quantified by the complex coefficient A_1. The real part describes astigmatism aligned with the horizontal and vertical axes, while the imaginary part corresponds to astigmatism oriented along the diagonal (45° and 135°) axes. This aberration causes the point spread function to become elongated along the corresponding directions.

Three-fold astigmatism (trefoil) is quantified by the complex coefficient A_2. It exhibits a threefold rotational symmetry and generates a three-lobed point spread function that is independent of over- or underfocus. The real and imaginary components of A_2 correspond to trefoil patterns aligned with peaks at 0° and 90°, respectively.

Comatic aberrations arise when optical rays entering the system at different positions and orientations relative to the optical axis focus at different axial locations, leading to an asymmetric distortion of the image [11]. This aberration depends on the direction of the incoming rays with respect to the central axis and lacks rotational symmetry. Second-order comatic aberrations are quantified by the complex coefficient B_2, which defines both the strength and orientation of the aberration. The real and imaginary components correspond to coma aligned along the horizontal and vertical directions, respectively. The θ term in the phase contribution introduces a compensating tilt that minimizes the effect of the comatic aberration [12].

Unlike comatic aberrations, spherical aberrations are rotationally symmetric and arise from rays focusing at different axial positions depending on their distance from the optical axis [11]. It is quantified by the real coefficient C_S, which defines the strength and direction (inner or outer) of the aberration. The θ term in the phase contribution adjusts the defocus such that the effect of the spherical aberration is minimised [12].

The characteristic impact of each aberration mode on the point spread function is illustrated in Figure 2. While additional aberration modes and shapes exist, this paper focuses on the modes discussed in this section, as they are the most significant and predominant for scanning electron microscopy.

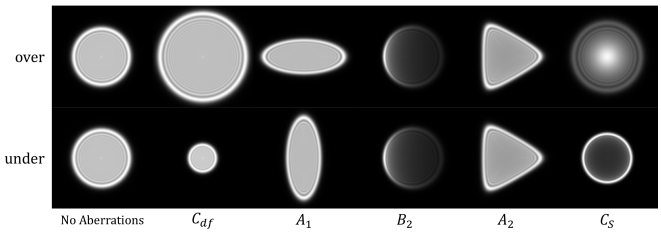


FIG. 2: Point spread functions of different aberration types under underfocus, nominal focus, and overfocus conditions, adapted from [7].

III. METHOD

A. PSF Extraction

As shown in Equation 2, the PSF $\Psi(x, y)$, when convolved with the intrinsic signal, results in the image produced by the SEM. In the Fourier space, this convolution turns into a multiplication. The Fourier transform of the PSF function $\Psi(x, y)$ is the Optical Transfer Function $\Phi(k, l)$, such that $\bar{\Phi}(k, l)$ is its complex conjugate.

$$I(k, l) = \Phi(k, l) \cdot S(k, l). \tag{6}$$

Assuming the same intrinsic signal S is used, a relation can be established between two images and their optical transfer functions.

$$\frac{\Phi_2}{\Phi_1} = \frac{I_2}{I_1} \tag{7}$$

With a through focus series of images, which consists of a set of images taken of the same sample through the same lens but at various defocus magnitudes, it is possible to compute the right hand side of Equation 7.

Firstly, the denominator is made real by multiplying both the numerator and the denominator by the complex conjugate of the focused image.

$$\frac{\Phi_2}{\Phi_1} = \frac{I_{df}}{I_f} \cdot \frac{\bar{I_f}}{\bar{I_f}} = \frac{I_{df} \cdot \bar{I_f}}{|I_f|^2} \tag{8}$$

To prevent the excessive propagation of noise, a Wiener deconvolution is applied by adding a constant to the denominator, shown in Equation 9 [13].

$$\Psi_{extracted} = \mathcal{F}^{-1}\left(\frac{I_{df} \cdot \bar{I_f}}{|I_f|^2 + w_c}\right) \tag{9}$$

where w_c is the Wiener constant. The larger the constant, the more the image is softened and the noise is reduced. However, this comes at the cost of detail in the extracted PSF.

According to Equation 7, the extracted PSF, $\Psi_{extracted}$, is equivalent to the fraction of optical transfer functions of the corresponding PSFs. Given that the defocus is known, it is possible to fit the remaining aberration coefficients such that the difference between the left and right hand sides of the equation is minimised.

Despite using a Wiener deconvolution, there is still a significant amount of noise present in the extracted PSFs. Some of this noise is particularly pronounced, such as sharp lines crossing the middle. This is shown in Figure 3.

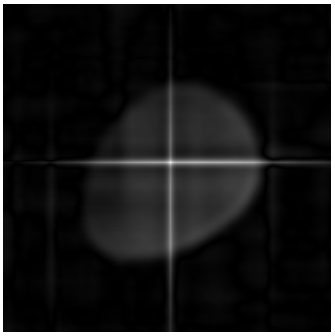


FIG. 3: PSF extracted from focused and defocused images using Equation 9 with a Wiener constant of 0.5.

These lines originate from the deconvolution, where the signals do not necessarily match at the edges. To reduce the resulting artifacts, a filtering window can be applied to the raw images. A classic example is the Hanning window, which filters the image with the Hann function (a half-period cosine that peaks at the center and tapers to zero at the edges) [14]. While effective at reducing edge discontinuities, this approach also removes a significant amount of information from the image, as illustrated in Figure 4.

Which Dataset Was Used For This? Others Are 57

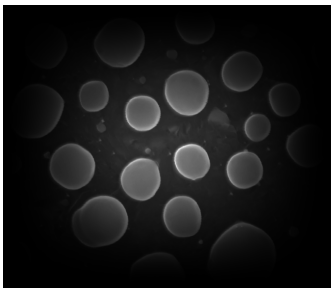
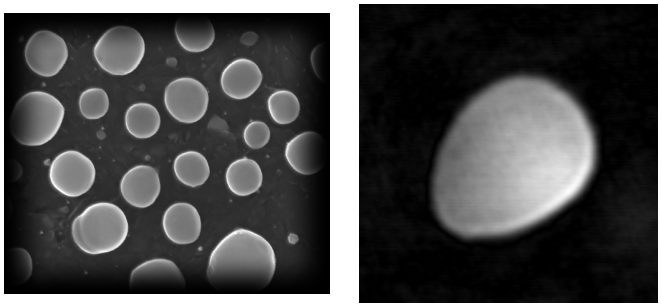


FIG. 4: Focused SEM image filtered with a Hanning window.

In contrast, a Tukey window only softens the edges of the image while preserving a selected portion of the central region [14]. For example, in Figure 5a, 30% of the image

is tapered, leaving the remaining 70% intact. Using this softened image, the PSF shown in **??** can be extracted.

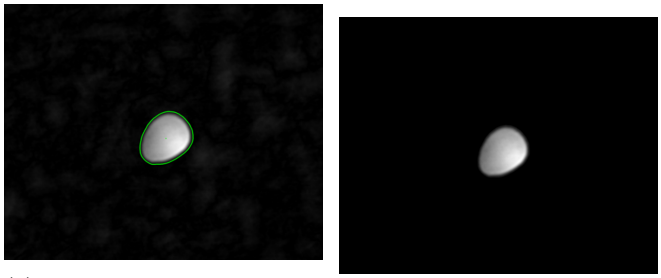


(a) Focused SEM image softened with a Tukey window.

(b) Extracted PSF after Tukey filtering.

FIG. 5: Effect of a Tukey window, with $\alpha = 0.3$, on PSF extraction.

Lastly, to remove the remaining noise, a contour is fitted around the PSF. For this, a threshold value of an intensity of 0.2 is chosen, such that the contour does not clip on any noise. This mask is then dilated by a margin of 15 pixels to preserve the shape and remaining details of the PSF, but also to ensure a more continuous change in the intensities. For everything outside the mask, the intensity is reduced to zero. An example of this procedure is shown in Figure 6.



(a) PSF fitted with a contour line (green) using an intensity threshold of 0.2, followed by a 15-pixel dilation.

(b) Denoised PSF after suppressing the intensities outside the mask.

FIG. 6: Example of the contour-based denoising procedure applied to the PSF.

B. PSF Generation

In order to compute the point spread function digitally, a discrete Fourier transform needs to be applied, together with relevant conversions between discrete pixel inputs and outputs. A two-dimensional discrete Fourier transform is defined as

$$F(k,l) = \frac{1}{MN}\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n) \cdot \exp\left[-2\pi i\left(\frac{k}{M}m + \frac{l}{N}n\right)\right], \tag{10}$$

where $M \times N$ is the resolution of the input into the Fourier Transform, $f(m,n)$ is the value of the function to be Fourier transformed with input (m,n), and $F(k,l)$ is the resulting Fourier Transformed function with new frequency coordinates (k,l).

Discretising Equation 5 and converting angular coordinates to pixel coordinates results in Equation 11,

$$U(x,y) = \frac{1}{n\theta_t\sqrt{\pi}}\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} \bar{P}(i-n_c, j-n_c) \cdot \exp\Big[-2\pi i\Big(\frac{x(i-n_c)}{n} + \frac{y(j-n_c)}{n}\Big)\Big], \tag{11}$$

where i and j are aperture pixel coordinates, θ_t is the maximum half opening angle in pixels, x and y are PSF pixel coordinates, $n \times n$ is the resolution of the aperture and PSF, and n_c is n halved and rounded down.

Performing the Fourier transform again on the PSF produces the optical transfer function (OTF) Φ. The electron beam itself, even if perfectly focused, has a finite width which must be taken into account for the production of the PSF. The electron beam can be approximated with a two dimensional Gaussian distribution, where the diameter containing 50% of the electron beam intensity can be estimated as variable $d_{50,geo}$. The standard deviation for the corresponding Gaussian distribution can be computed through Equation 12.

$$\sigma = \frac{d_{50,geo}}{\sqrt{\ln(256)}} \tag{12}$$

This Gaussian distribution can be applied to the PSF through a convolution in regular space, or through a multiplication with the relevant OTF in the frequency space.

As shown in Equation 7, the reproduction of the generated PSF also requires a Wiener deconvolution to reduce noise and prevent division by zero. This means that the final generated PSF is computed using the relevant focused and defocused OTFs Φ_f and Φ_{df} respectively through Equation 13 [13].

$$\Psi_{generated} = \mathcal{F}^{-1}\left(\frac{\Phi_{df}\cdot\bar{\Phi_f}}{|\Phi_f|^2 + w_c}\right) \tag{13}$$

C. Seidel Aberrations Extraction

Once the PSF has been extracted from the SEM images and the dominant noise components have been removed, as described in section III A, an optimization procedure is employed to estimate the Seidel aberration coefficients. This procedure requires an initial guess for each coefficient, along with appropriate bounds. In this work, all Seidel aberrations up to second order, as well as spherical aberration, are included in the model. In addition, $d_{50,geo}$, zpf, and θ, are treated as additional optimization parameters, since they cannot be determined accurately prior to optimization.

Initial estimates for the defocus term, C_{df}, and for the half-opening angle, θ, are obtained from approximate analytical relations, providing a physically motivated starting point for the optimization. C_{df} is estimated based on Equation 14, where UHR_f and UHR_{df} are the objective lens currents in ampere of the focused and defocused image, in the ultra-high resolution mode. UHR_{cal} is a constant of the SEM that converts the change in current to a defocus in micrometers.

$$C_{df} = (UHR_{df} - UHR_f) \cdot UHR_{cal} \tag{14}$$

From the C_{df} estimate, an initial approximation of θ (in radians) can be obtained using Equation 15, based on the reasoning illustrated schematically in Figure 7. Here, scale denotes the physical distance per pixel (in meters), and r_{PSF} represents the average PSF radius in pixels. This radius is defined as the equivalent radius of a circle whose area matches that of the PSF, computed by summing all pixels within the PSF contour with intensity above 0.2.

$$\theta = \arctan\left(\frac{r_{PSF} \cdot L_{pix}}{\mid C_{df} \mid}\right) \tag{15}$$

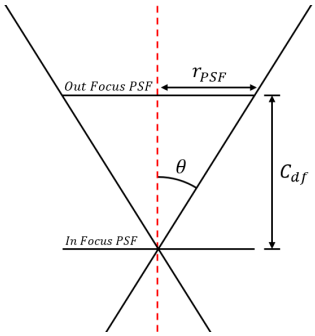


FIG. 7: Schematic illustration of the SEM geometry used to estimate the half-opening angle θ from the defocus term, C_{df}, and the radius of the out-of-focus PSF, r_{PSF}.

In contrast, zpf and $d_{50,geo}$ are initialized to 1.10 and 10 nm, respectively, based on repeated experience with comparable SEM systems. All remaining aberration coefficients are initialized to zero, representing an aberration-free system at the start of the optimization.

For all parameters, bounds are imposed to restrict the search space to physically plausible values, and to ensure stable and efficient convergence of the optimization algorithm. These bounds are summarized in Table II. The zero-padding factor zpf is technically limited between 1 and 15 [] , but based on prior experience, it is typically constrained near 1.10 to ensure the generated PSF remains physically meaningful. The same rationale applies to $d_{50,geo}$. The bounds for θ and C_{df} are narrower, reflecting the fact that their initial estimates, θ_{est} and $C_{df,est}$, are already fairly accurate. In particular, since the C_{df} approximation tends to overestimate the true value, a smaller lower bound is used. Finally, the bounds for the Seidel aberration coefficients are chosen based on empirical experience and physical considerations.

Source

TABLE II: Parameter bounds for the geometric blur, zero-padding factor, half-opening angle, and the Seidel aberrations used in the optimization procedure.

Parameter	Lower Bound	Upper Bound
$d_{50,geo}$	1.0 nm	20.0 nm
zpf	1.09	1.11
θ	$0.95 \cdot \theta_{est}$	$1.05 \cdot \theta_{est}$
C_{df}	$0.75 \cdot C_{df,est}$	$1.05 \cdot C_{df,est}$
A_{1r}	-50.0 μm	50.0 μm
A_{1i}	-50.0 μm	50.0 μm
B_{2r}	-200.0 μm	200.0 μm
B_{2i}	-200.0 μm	200.0 μm
A_{2r}	-200.0 μm	200.0 μm
A_{2i}	-200.0 μm	200.0 μm
C_3	-30.0 mm	30.0 mm

With the initial parameter guesses and bounds defined, the final step before running the optimization algorithm is to construct the residual function. The residual is defined as the difference between the extracted PSF and the PSF generated from the current set of model parameters. A key challenge in this comparison is ensuring that the two PSFs are directly comparable. First, the images must be spatially aligned, as even small offsets can produce large, non-physical residuals. To mitigate this effect, both PSFs are centered using their measured centroids prior to computing the residual.

Once alignment is achieved, the PSFs must be scaled such that the pixels from the extracted PSF and the generated PSF have the same physical size. The size of each pixel L_{pix} in the generated PSF is computed using Equation 16.

$$L_{pix} = \frac{\lambda}{2\theta \cdot \text{zpf}}, \tag{16}$$

where λ is the relativistic De Broglie wavelength of the electrons, θ is the half opening angle of the aperture and zpf is the zero padding factor used in generation. The number of pixels of the generated PSF is determined based on the ratio between the pixel size of the extracted PSF and the generation pixel size. The subsequent generated PSF is then downscaled to match the pixel size of the extracted PSF, enabling the residual to be computed.

Now that everything is defined, the actual parameters can be extracted from the PSF. This is accomplished using a least squares approach in combination with a trust-region reflective (TRF) algorithm. TRF is selected because it is particularly effective for nonlinear, multi-dimensional, bounded optimization problems. The algorithm constructs a quadratic approximation of the objective function and defines a so-called trust region within which this approximation is considered reliable. A step is then taken within this region to minimize the model, and, consequently, the objective function. After several iterations, the model converges to a minimum, yielding the retrieved parameters. [15, 16]

IV. RESULTS

A. Selection of the Optimal Wiener Constant

As described in section III A, the Wiener constant plays a role in reducing noise during the deconvolution of the extracted PSF. Choosing an appropriate value for this constant requires balancing two competing factors. A lower Wiener constant preserves fine features, especially spherical aberration in underfocus, but increases noise, resulting in a higher MSE. Conversely, a higher constant suppresses noise and lowers the MSE, but this comes at the cost of smoothing out PSF features, as illustrated in Figure 8.

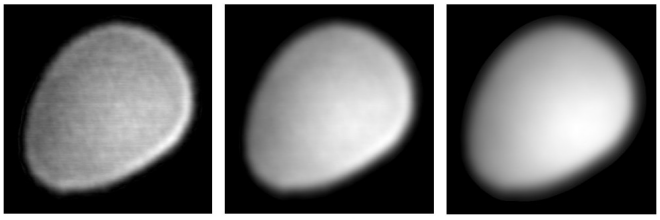


FIG. 8: PSFs extracted from the same input images using different Wiener constants (left to right: 0.1, 0.5, and 2).

Several Wiener constant values were evaluated, and the resulting data is shown in Figure 9 and Figure 10. The trends described above are clearly observable. A value of 0.5 was ultimately selected as a compromise, balancing feature preservation with error reduction.

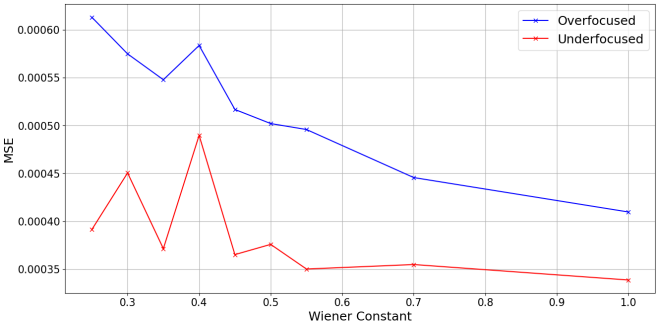


FIG. 9: Mean squared error of the deconvolved PSF as a function of the Wiener constant. Overfocus is shown in blue and underfocus in red.

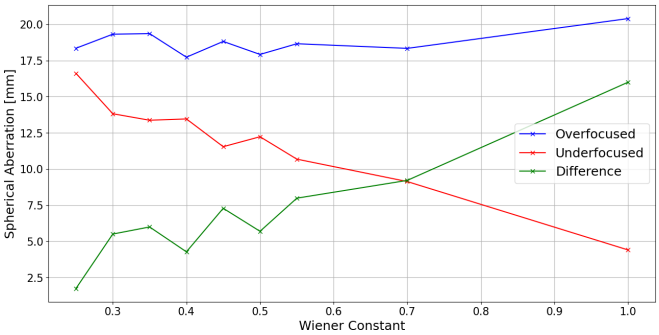


FIG. 10: Recovered spherical aberration of the deconvolved PSF as a function of the Wiener constant. Overfocus is shown in blue, underfocus in red, and the difference between them in green.

B. Method Verification

To verify if the produced code can correctly extract the aberrations from an PSF two different tests were done. The objective in both tests is the same, however the source of the PSF's differs in both tests. The first test uses self-generated PSF's. The aberrations of these PSF's are precisely known. The downside of this test is that the PSF's do not have any noise. Additional the generated PSF's could be non physical. The second test uses PSF's that are extracted from the in and out focus images from the SEM. This test contains only physical PSF's, because the PSF's are extracted from real data. The downside is that the values of the aberration coefficients are not known, so it is difficult to quantify how precise the code is.

1. *Verification using self-generated PSFs*

The aberration coefficients for the self-generated PSF's were chosen based on extracted data from real PSF's, to ensure that the PSF's were physically meaningful. The self generated PSF can be seen on the left in Figure 11.

The PSF generated by the code based on the extracted coefficients can be seen in the middle of Figure 11. In table III you can see the errors per coefficient.

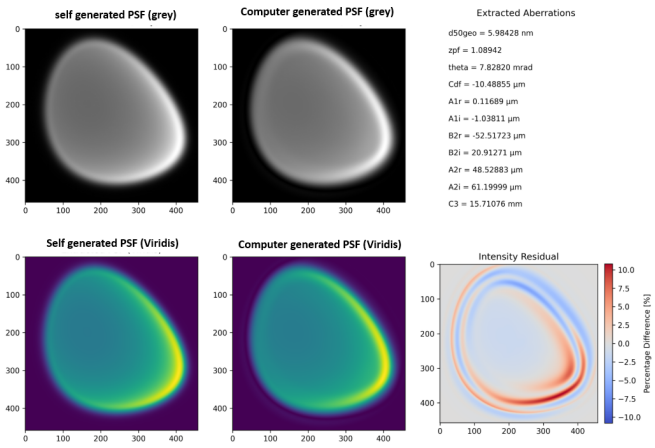


FIG. 11: self-generated PSF test

Parameter	actual value	extracted value	Error (%)
$d50_{geo}$	8.0 nm	5.98428 nm	−25.20
zpf	1.10	1.08942	−0.96
θ	8.0 mrad	7.82820 mrad	−2.15
C_{df}	−10.0 μm	−10.48855 μm	+4.89
A_{1r}	0.10 μm	0.11689 μm	+16.89
A_{1i}	−1.00 μm	−1.03811 μm	+3.81
B_{2r}	−58.0 μm	−52.51723 μm	−9.45
B_{2i}	26.0 μm	20.91271 μm	−19.57
A_{2r}	52.0 μm	48.52883 μm	−6.68
A_{2i}	57.0 μm	61.19999 μm	+7.37
C_3	18.0 mm	15.71076 mm	−12.72

TABLE III: SEM and aberration parameters: initial values, updated values, and relative errors

2. *Verification using extracted PSFs*

This verification test is mainly used to see if the code can handle the added noise from real data. The PSF that was used in this test was extracted from image set 55 (see appendix).
The PSF generated based on the extracted aberrations can be seen in Figure 12. It can be seen that the generated PSF's have similar shape and intensity as the extracted PSF. The intensity of the residuals shows that the error is almost of the same magnitude as the error in the self-generated PSF test.

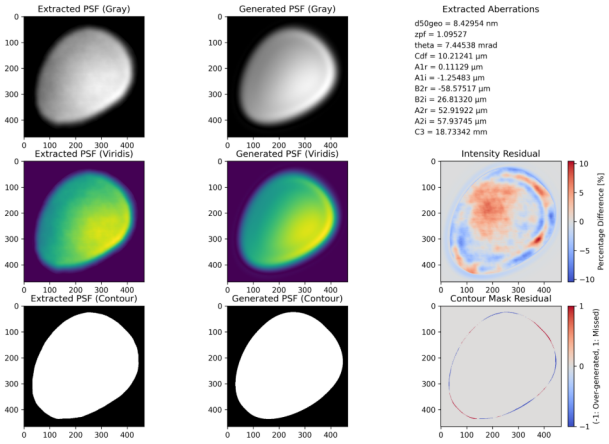


FIG. 12: extracted PSF test

C. Parameter Retrieval

move second verifcation part, i.e. extracted PSF retrieval here

V. DISCUSSION

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VI. CONCLUSION

In this work, a method was developed to quantify Seidel aberration coefficients in scanning electron microscopy by fitting numerically generated point spread functions to probe shapes extracted from through-focus SEM images. The approach combines Fourier-optical PSF modelling with Wiener-based deconvolution and a constrained nonlinear optimization procedure, forming a physically motivated framework for aberration retrieval.

Verification using self-generated PSFs showed that the algorithm can recover known aberration coefficients, including spherical aberration, defocus, astigmatism,

and coma, with reasonable relative errors. When applied to experimentally extracted PSFs, the method remained stable despite the presence of noise and deconvolution artefacts. In these cases, the generated PSFs closely resembled the extracted probe shapes in both structure and intensity distribution, indicating that the approach is suitable for realistic SEM data.

A consistent difference was observed between underfocused and overfocused probe shapes. Underfocused PSFs generally resulted in more reliable fits, while overfocused PSFs were more affected by edge artefacts introduced during PSF generation and deconvolution. This highlights the importance of careful preprocessing, windowing, and the choice of the Wiener constant in order to obtain physically meaningful results.

Although the method successfully demonstrates the feasibility of aberration quantification from SEM images, several limitations remain. Further improvements could be achieved by refining the physical modelling of the PSF, particularly at larger aperture angles.

Overall, this work provides a solid foundation for quantitative aberration analysis in scanning electron microscopy. With further development, including the use of larger datasets and automated fitting strategies, the approach could contribute to practical tools for monitoring and evaluating aberration correction in high-resolution SEM systems.

APPENDIX

A. Iteration Series

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B. Verification Series

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